Beyond Optimistic Planning: When Does Learning Stochastic User Dynamics Make Recommendation Planning Better?

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Abstract

The MF-MDP formulation casts top-K retrieval as planning over the implicit-ALS posterior: the state is the fold-in posterior (P, u), the transition is a closed-form rank-one update that *optimistically* assumes every recommendation is accepted, and the reward mixes cosine relevance with a posterior-alignment term. We reproduce this method and show that, on MovieLens-1M under the leave-last-n protocol, the static baseline is reproduced (Recall@10 0.0704 vs. 0.0694 reported) but the reported planning gains are not. A leverage analysis explains why: the quantity driving the reward's alignment term is bounded by $\approx d/n_{\text{items}}$ for spread embedding geometries and correlates with item popularity rather than user relevance, leaving the planning layer quality-inert. We then ask whether *learning stochastic user responses* fixes this. In two controlled environments with known generative dynamics and exact oracles we show that feedback-conditioned stochastic planning reaches the oracle and beats the optimistic planner by $+0.122$ (paired bootstrap CI95 $[0.115, 0.129]$) when the state representation is faithful, and does not differ from it (-0.004, CI95 $[-0.010, 0.002]$) when the state is a compressed MF posterior. The benefit of deeper planning is governed by the same bottleneck: regret falls 5–15$\times$ from depth-2 to full-horizon planning once the model is reasonable, and response-model estimation error stops being binding once the model saturates. On KuaiRand-1K, graded engagement (watch-ratio) is predictable and transfers to randomized exposure (AUC 0.839 $\rightarrow$ 0.826), while binary click prediction is policy-bound and temporally miscalibrated (ECE 0.234). Our answer to "when does learning a stochastic recommendation environment make planning better?" is: only when the state representation carries the dynamics — otherwise the transition model's sophistication is irrelevant.

1 Introduction

A recommendation is never an isolated event. A consumed item changes the user, and therefore what should be retrieved next; a poor pick can end the session or push the user into an irrelevant region of the catalog. Sequential recommendation is naturally a Markov decision process: the state describes the user's current context, the action is the recommended item, the reward is the user's feedback, and the transition encodes how the state evolves. Exploiting this structure for *candidate generation* — quickly selecting a small set of promising items from a large catalog — is the goal of the starting paper [15], which builds an MDP directly on the implicit-ALS posterior. The user state is the pair (P, u) where $P = A^{-1}$ is the inverse precision of the fold-in least-squares problem, actions are catalog items with fixed embeddings, and the transition is a closed-form Sherman–Morrison rank-one update [13] that folds the recommended item in as *accepted* positive feedback — an **optimistic** dynamics model. The reward combines a cosine relevance term with a posterior-alignment term driven by the leverage $\ell = v^\top P v$, and the compared planners span static retrieval, one-step lookahead (Plan-1), and horizon-K MCTS (Plan-K).

The optimistic transition is, however, a misspecified model of the user: real users skip, partially consume, dislike, or leave. Whether planning over such a posterior helps at all — and whether replacing the optimistic assumption with a *learned stochastic response model* changes the answer — is an open and falsifiable question. It cannot be answered from logged-data metrics alone, because the counterfactual dynamics are unobservable: responses to unexposed items are unknown, and the logging policy confounds what is visible.

What exists. Learned world models for recommendation are an active area. MCRL [9] learns reward and transition structure with contrastive objectives; ROLeR [16] and R3S [14] refine reward models and uncertainty penalties; KuaiSim [17], RecSim NG [4], and the KuaiEnv of CIRS [2] provide hand-built simulators; a recent session-level REINFORCE method [10] simulates user feedback at test time; WMG-RL [8] distills a frozen user engagement world model into a policy. Model-based RL contributes the planning machinery: MuZero [12], Sampled MuZero [6] for large action spaces, Stochastic MuZero's chance nodes [1], MOReL's pessimism under model error [7], and the value-equivalence principle [3] that justifies planning over a compressed, value-oriented state.

What is missing. None of these works measures, against a *known* true environment, whether feedback-conditioned stochastic dynamics create planning value *specifically over the analytic MF posterior* — the representation used in production retrieval stacks via iALS [5, 11]. The starting paper evaluates only on logged data; the learned-model approaches evaluate inside their own models or on offline proxies. The contribution of the transition model itself is therefore confounded.

Our proposal. We keep the analytic posterior and replace only the optimistic assumption. A learned stochastic response model $p_\phi(y \mid s, a)$, trained on decision-time features, feeds exact feedback-conditioned weighted fold-in transitions (α, β), and a chance-node lookahead planner evaluates actions under the resulting Bellman equation. All claims are validated in controlled environments with known dynamics and exact oracles, at matched compute budgets, with paired bootstrap confidence intervals.

Contributions.

- A test-verified reproduction of the MF-MDP pipeline and a mechanistic explanation of its inertness on MovieLens-1M: the alignment-driving leverage is a popularity proxy bounded by $\approx d/n_{\text{items}}$.
- A controlled-environment answer to when stochastic planning helps: faithful state $\rightarrow$ near-oracle gains; compressed MF posterior $\rightarrow$ no effect.
- An error-versus-depth analysis: deeper planning cuts regret sharply once the model is reasonable; model error stops binding once the response model saturates.
- A KuaiRand-1K validation: graded engagement transfers to randomized exposure; binary click does not.

2 Problem statement

Plain form. At each step a recommender shows one item a; the user responds stochastically with $y \in \{\text{engage}, \text{skip}\}$ and may terminate; the response changes the user's future preferences. We ask which planning model — optimistic or stochastic — produces better *sequences* of recommendations.

Table 1: Closest work vs. ours (full table in the repository).

Work	State	Dynamics	Planning
MCRL 2023	neural seq	contrastive	none
ROLeR 2024	neural	world model	none
R3S 2025	neural	world model	none
KuaiSim 2023	simulator	hand-crafted	evaluation
RecSim NG 2021	ecosystem	probabilistic program	—
CIRS 2022	latent interest	counterfactual causal	—
Session-REINFORCE 2026	two-tower	feedback model	rollout scaling
WMG-RL 2026	LLM/UEWM	learned env model	none
This work	**MF posterior**	**learned response + exact fold-in**	**chance-node lookahead**

Formal statement. Let s_t be a representation of the user's history, a_t an item from the catalog, y_t the response, d_t termination, and r_t the immediate utility. A planner maps states to actions. The objective is the expected discounted return under the *true* transition T^*:

$$J(\pi) \;=\; \mathbb{E}_{\pi,T^*}\left[\sum_{t=0}^{H-1}\gamma^t r_t\right] \quad\longrightarrow\quad \pi^* \in \arg\max_{\pi\in\Pi} J(\pi). \tag{1}$$

The internal training criterion is the negative log-likelihood of observed responses (and, for value heads, Monte-Carlo return-to-go regression under a *named* policy). External criteria are $J(\pi)$ in the true environment, regret against an exact oracle, Recall@10 / nDCG@10 on logged data (secondary only), and latency. Constraints: offline data support, matched compute budgets, documented assumptions. The model class: logistic/MLP response heads, analytic weighted fold-in transitions, ridge/MLP value heads, and chance-node lookahead search.

Labeling contract. Four concepts are kept separate throughout: (i) the response distribution p_ϕ; (ii) the utility mapping $U(y)$; (iii) the logging-policy value V_{logging}, estimated from logged trajectories; (iv) the target-policy value V^π, which we do not claim to estimate on real data. Monte-Carlo returns from logs are always reported as logging-policy value.

3 Related work

Table 1 summarizes the closest lines; the full table is in the repository (`docs/LITERATURE_REVIEW_PHASE5.md`). The defensible gap is the controlled-environment measurement of optimistic-vs-stochastic planning over the analytic MF posterior, with error-vs-depth accounting. We do not claim novelty for learned world models, reward or value learning, or MCTS per se.

4 Method

4.1 The MF-MDP and the optimistic baseline

For a user history H, the fold-in solves $\min_u \sum_i c_i(p_i - u^\top v_i)^2 + \lambda\|u\|^2$ with $A = \lambda I + \sum_i c_i v_i v_i^\top$, $b = \sum_{i\in H} c_i v_i$, $u = A^{-1}b$. The state is $s = (P, u)$ with $P = A^{-1}$. The optimistic transition folds the chosen item a in as a unit-confidence positive:

$$z = Pv, \quad \ell = v^\top P v, \quad P^+ = P - \frac{zz^\top}{1+\ell}, \quad u^+ = u + \frac{z(1-u^\top v)}{1+\ell}, \tag{2}$$

with self-score shift $\Delta s(a) = \frac{\ell}{1+\ell}(1 - u^\top v)$ and reward $r(s,a) = \text{sim}(u, v_a) + \eta/(1 + |\Delta s(a)|)$. We verified all closed forms against direct recomputation (inverse and weighted least squares).

4.2 Reproduction findings that shape the method

Our reproduction (Section 6.1) shows the static baseline is reproduced but the planning gains are not. The mechanism, established by a dedicated diagnostic phase, is structural: for spread embedding geometries the leverage obeys $\ell \lesssim d/n_{\text{items}}$ (verified via the trace identity $\lambda \operatorname{tr}(P) + \sum_i \ell_i + \alpha \sum_{i \in H} \ell_i = d$), and ℓ correlates with item norm and popularity (Spearman up to 1.0 / 0.96) far more than with user relevance. The alignment term is therefore inert where embeddings are good and popularity-sorted where it is active. Learned reward variants (response distribution $\times$ fixed utility; direct expected-utility regression) lose to the paper heuristic on MovieLens-1M, whose only labels are 4-vs-5-star ratings on consumed items (response AUC 0.62).

4.3 The stochastic model

The response model $p_\phi(y \mid s, a)$ is a logistic or small-MLP head on decision-time features (cosine, leverage, self-shift, item norm, train-period popularity, state summaries); a termination hazard head models churn. Transitions are *exact weighted fold-ins*: for response type with confidence increment α_w and target increment β_w,

$$P^+ = P - \frac{\alpha_w z z^\top}{1 + \alpha_w \ell}, \qquad u^+ = u + \frac{z(\beta_w - \alpha_w u^\top v)}{1 + \alpha_w \ell}, \tag{3}$$

which we proved and tested to equal the exact re-fold of the extended history for $(\alpha_w, \beta_w) = (\alpha, 1 + \alpha)$ on engagement. Planning uses chance-node lookahead,

$$Q(s, a) = \mathbb{E}_{y \sim p_\phi} \left[r(s, a, y) + \gamma (1 - h(s, a))\, V(s') \right], \tag{4}$$

with V either the exact oracle (Env-S), a bounded-candidate closed form, or a learned leaf value.

4.4 Controlled environments (Experiment A)

Env-S (exact oracle). Twelve discrete states (interest mode $\times$ fatigue), ten items with documented affinities and qualities. Engagement $p_i(s) = \sigma(4(m_i[s] - 0.5) + q_i)$; reward $q_i m_i[s](1 - 0.35f)$ on engagement; assimilation to the item's dominant mode w.p. $\rho = 0.6$; termination hazard $\sigma(-1.5 - 4r)$. Handcrafted high-quality off-mode items create greedy-vs-optimal tension. The exact oracle is backward induction over the 12-state MDP.

Env-C (approximate oracle). Continuous true state $\theta_t \in \mathbb{R}^{16}$ with assimilation/drift dynamics, latent per-step response noise, termination hazard on mismatch; iALS factors are trained on logged trajectories; planners see only the MF posterior (P, u) and use learned response/termination heads with exact weighted fold-ins. The oracle is a high-budget MCTS in the true environment (labeled approximate).

5 Experimental setup

MovieLens-1M under leave-last-1 ($n_{\text{test}} = 1$) for the reproduction; MF hyperparameters are unreported in the starting paper and were selected on a validation sweep (documented in `docs/DECISIONS.md`). Env-S: 5 seeds, 4000 rollouts per policy, exact DP oracle; stochastic components trained on trajectories from ε-greedy and uniform behavior policies. Env-C: 5 seeds, 200 evaluation users per seed, true-environment $J(\pi)$. KuaiRand-1K: chronological split (standard period 1 $\rightarrow$ standard period 2), decision-time-only features, the randomized-exposure file as an unbiased probe. Reported metrics: $J(\pi)$, regret vs oracle, Recall@10/nDCG@10, latency; AUC / log-loss / Brier / ECE for response models.

Table 2: MovieLens-1M LLN reproduction (6035 test users, 3 seeds): ours vs. the paper.

Method	Ours R@10	Paper R@10	Ours nDCG	Paper nDCG
Static (dot)	**0.0704**	0.0694	0.0343	0.0567
Static cosine (ablation)	0.0696	—	0.0342	—
Plan-1 (cos)	0.0694	0.0762	0.0341	0.0612
Plan-1 (dot ablation)	0.0704	0.0630	0.0343	0.0513
Plan-K (subset, 1 seed)	0.0685	0.0764	0.0343	0.0611

Table 3: Env-S ablation (5 seeds, 4000 rollouts, matched depth-H budgets): true-environment return $J(\pi)$. Stochastic − optimistic = +0.122, CI95 $[0.115, 0.129]$.

Planner	$J(\pi)$ mean	$J(\pi)$ std
Greedy (myopic)	0.559	0.002
Optimistic depth-H (paper T0)	0.523	0.012
Stochastic depth-H (ours)	**0.645**	0.007
Oracle (exact DP)	0.651	0.007

6 Results

6.1 Reproduction and diagnostics (MovieLens-1M)

Table 2: Static(dot) is reproduced within the paper's variance; the planning gains are not. The diagnostics phase established the mechanism (leverage as a popularity proxy; alignment inert or harmful; nine geometry variants without a planning gain). A candidate-generation bug found during the reward-semantics audit (consumed items leaking into ANN pools when $l \geq n_{\text{unconsumed}}$) was fixed and regression-tested; the earlier "popularity amplification" interpretation was retracted in a dated addendum — the paper's alignment term *decreases* in leverage.

6.2 Controlled environments

A1 (planning correctness). Greedy differs from the DP-optimal policy ($J(\pi)$ gap 0.084, CI95 $[0.069, 0.099]$); chance-node MCTS converges to the DP choice as the simulation budget grows (residual disagreements are statistically tied states). This cross-check caught a bug in our first DP oracle (it ignored assimilation failure); the corrected oracle is Monte-Carlo verified.

A3 (Env-S ablation). Table 3: the stochastic planner nearly matches the exact oracle (regret 0.006) and beats the optimistic planner decisively. The learned response model recovers the environment (AUC 0.75, ECE 0.012; assimilation estimate $\hat{\rho} = 0.599$ vs. true 0.6).

A4 (Env-C ablation). Table 4: over the compressed MF posterior, stochastic and optimistic planners do not differ; both sit far below the true-state greedy. The bottleneck is the representation (iALS alignment with the true factors ≈ 0.6), not the transition model.

H3 (depth × noise). Table 5: regret falls 5–15× from depth-2 to depth-6 at every noise level; estimation error (training-set size) further reduces regret at depth 6, but the response model saturates quickly — the transition structure is exactly known here, so no "depth hurts" regime appears.

6.3 KuaiRand-1K

Table 6: graded engagement (watch-ratio ≥ 0.8 on clicks) is predictable and transfers to randomized exposure; binary click is policy-bound and temporally miscalibrated (ECE 0.234 on the

Table 4: Env-C ablation (MF posterior; 5 seeds, 200 users each): stochastic − optimistic = −0.004, CI95 $[-0.010, 0.002]$.

Planner	$J(\pi)$ mean	$J(\pi)$ std
Greedy MF (no planning)	0.171	0.039
Optimistic 2-step	0.217	0.059
Stochastic 2-step	0.213	0.054
True-state greedy (reference)	0.685	0.082

Table 5: Env-S regret vs exact oracle by planning depth and response-noise level (3000 training trajectories; full grid in the repository).

Response noise σ	depth 2	depth 6	depth 2 (250 traj)	depth 6 (250 traj)
0.5	0.110	0.013	0.110	0.025
1.5	0.124	0.001	0.124	0.026
3.0	0.134	0.009	0.134	0.041

chronological test) — direct evidence of temporal drift for calibration-sensitive uses.

7 Limitations

Env-S and Env-C are handcrafted; their contrast indicates the role of the state representation but is not a strict single-variable isolation. Simulator results are diagnostics, not evidence of real-user improvement. MovieLens-1M lacks exposure labels; KuaiRand propensities are not verified, so no off-policy evaluation claims are made. V_{logging} (logging-policy value) is never V^{π}; target-policy value remains unimplemented. The KuaiRand response model uses basic features; the 3.4GB statistic feature file is unused.

8 Conclusion

The answer to our research question is conditional: **learning stochastic user dynamics makes model-based planning better than optimistic or greedy recommendation if and only if the state representation carries the dynamics.** When the state is faithful, a learned response model plus exact feedback-conditioned transitions recovers the oracle; when the state is a compressed MF posterior, the transition model's sophistication is irrelevant. This shifts the design question from "how to plan deeper" to "what state to build". All results are reproducible with `python run_experiment.py`; the experiment log records every run with metrics.

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Table 6: KuaiRand-1K response models (chronological split; 5M training rows). Popularity-only baselines: AUC 0.51–0.54.

Target	Test AUC	Test ECE	Rand AUC	Rand ECE	Base rate (test)
click	0.607	0.234	0.515	0.438	0.377
long_view	0.627	0.104	0.542	0.366	0.261
wr80 on clicks	**0.839**	**0.043**	**0.826**	0.229	0.559

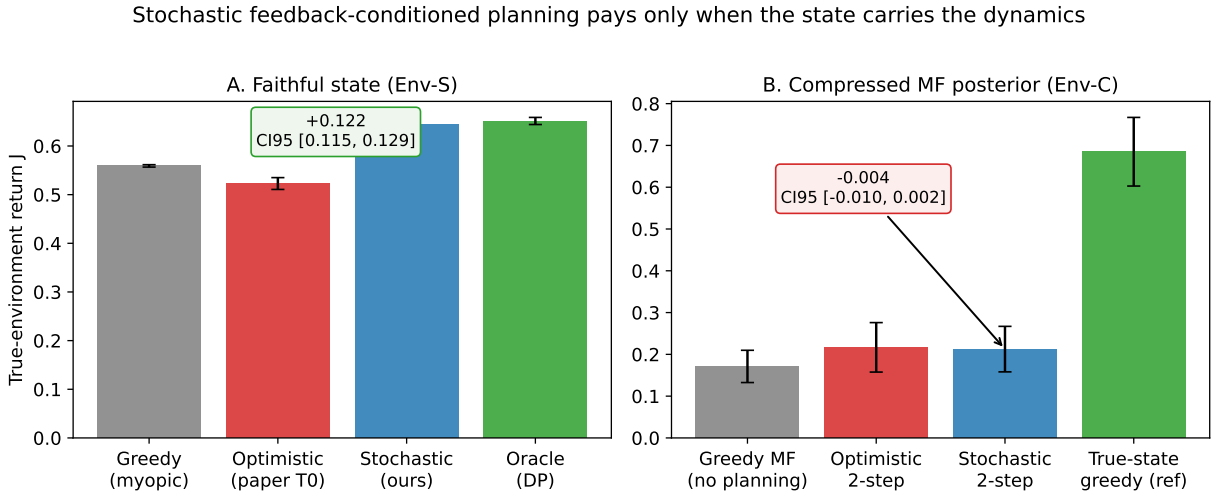


Figure 1: The project's main message: stochastic feedback-conditioned planning pays only when the state carries the dynamics. Left (faithful state, Env-S): stochastic ≈ oracle, +0.122 over optimistic. Right (compressed MF posterior, Env-C): no difference.

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